# Memory Organization : Secondary Memory

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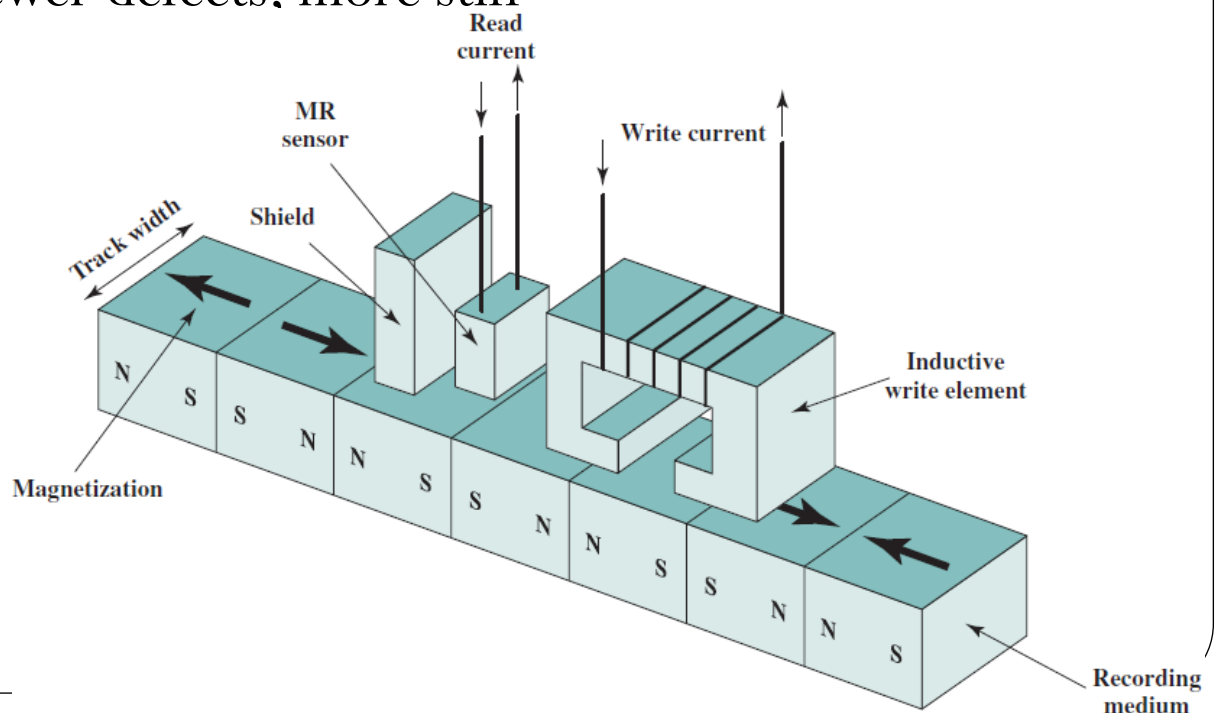
# External Memory

- Magnetic disks – fundamental external memory
- A disk is a circular platter constructed of nonmagnetic material **(substrate)** coated with a magnetizable material
  - Aluminium or aluminum alloy substrate
  - Glass substrate – uniform surface with fewer defects, more stiff
- The direction of magnetic field represents 0 or 1
  - If one direction is chosen to represent 0 then the opposite is direction is 1



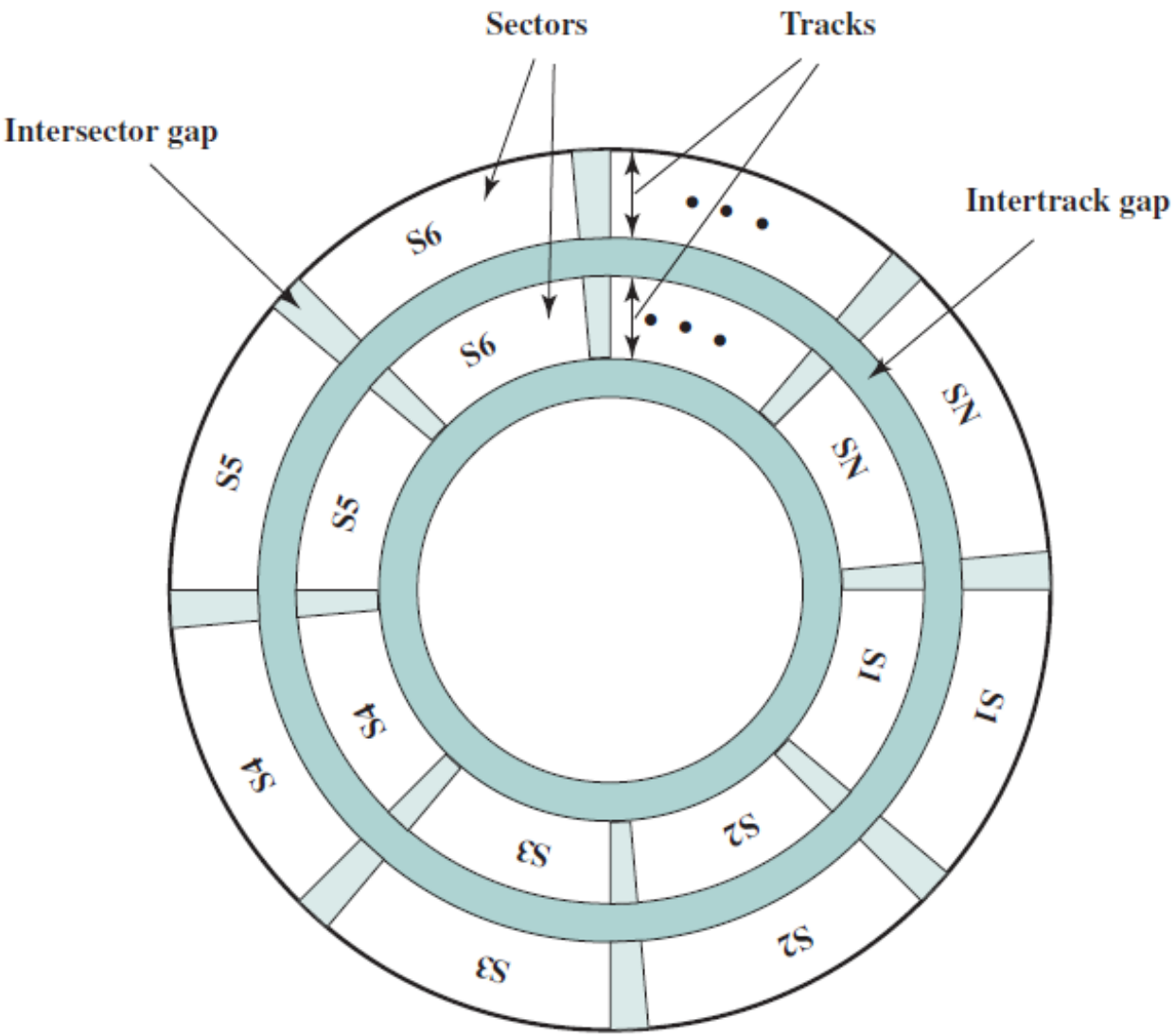
# Magnetic Disk – Read/Write

- Data are recorded on and later retrieved from the disk via a conducting coil named the **head**
  - During a read or write operation, head is stationary while the platter rotates beneath it
- **Write** mechanism exploits the fact – electricity flowing through a coil produces a magnetic field
  - Magnetic patterns are recorded on the surface below, with different patterns for +ve and –ve currents
- **Read** mechanism exploits the fact – magnetic field moving relative to a coil produces an electrical current in the coil
  - When surface of the disk passes under the head, it generates a current of the same polarity as the one already recorded on it

# Magnetic Disk – Data Organization

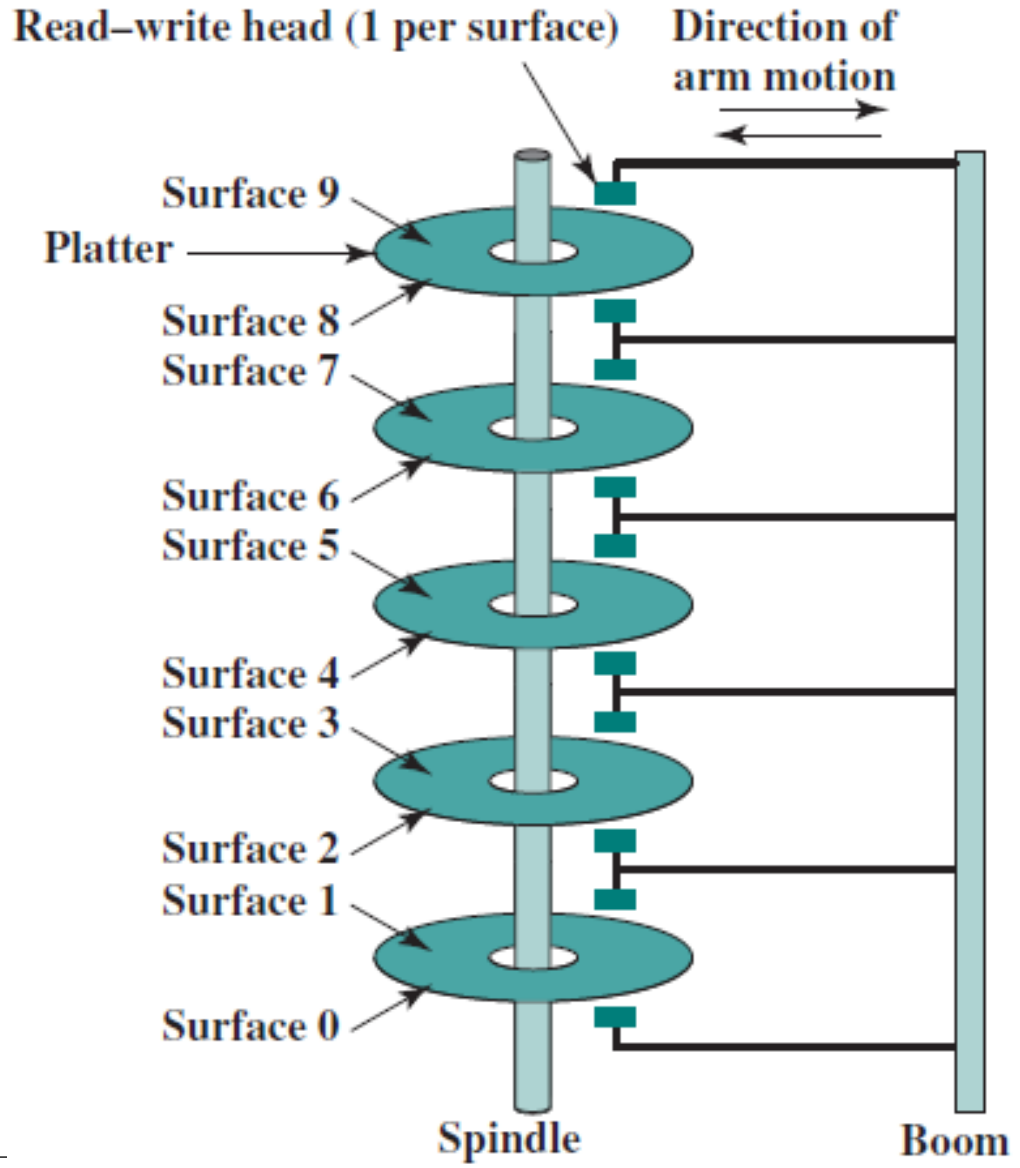
- Data is organized on the platter in a concentric set of rings, called **tracks**
  - Tracks have same width as the head; 1000s of tracks per surface
- Adjacent tracks are separated by **gaps**
  - To prevent/minimize, errors due to misalignment of the head or interference of magnetic fields
- Each track is organized into **sectors** (same or variable size)
  - 100s of sectors per track; data is transferred to and from disk in sectors; 512B is usually the size of a sector
  - adjacent sectors are separated by intratrack (intersector) gaps

# Magnetic Disk



# Magnetic Disk - Characteristics

- **Multiple platters**
  - stacked vertically a fraction of an inch apart
  - multiple arms, movable head, with one r/w head per surface
  - all heads mechanically fixed so that all are at the same distance from the centre of the disk and move together
  - set of all the tracks in the same relative position on the platter is referred to as a **cylinder**



# Magnetic Disk - Characteristics

- **Fixed head** or **movable head**
  - Fixed head: one r/w head per track. All heads are mounted on a rigid arm that extends across all tracks
  - Movable-head: there is only one r/w head. Head is mounted on an movable arm that can be positioned above any track
- **Removable** or **non-removable disk**
  - Non-removable disk – permanently mounted; hard-disk
  - Removable disk – can be removed/replaced with another disk
- **Double sided** or **single-sided**
  - Double sided: magnetizable coating applied on both sides ; 2 surfaces per disk

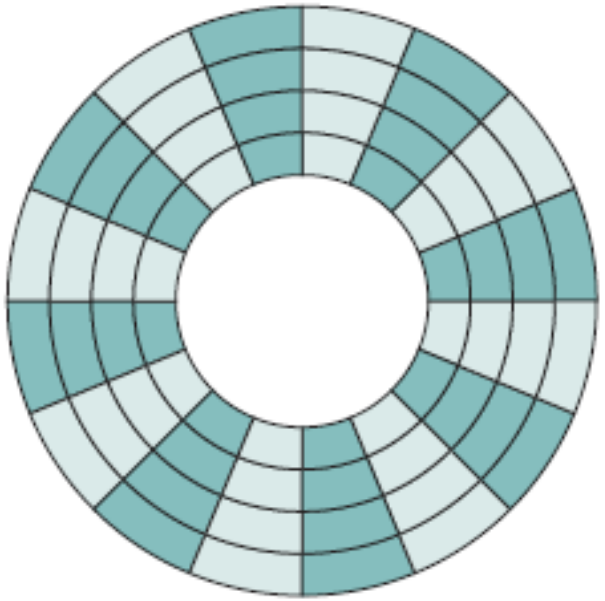
# Magnetic Disk – Data Organization

- A bit near the centre of a rotating disk travels past read–write head slower than a bit on the outside
  - Compensate for the variation in speed – **Increase spacing between bits of information recorded in segments of the disk** – Information thus scanned at the same rate by rotating the disk at a fixed speed, known as the **constant angular velocity** (CAV)
- Disadvantage: is that the amount of data

that can be stored on the long outer tracks

is same as what can be stored on the short

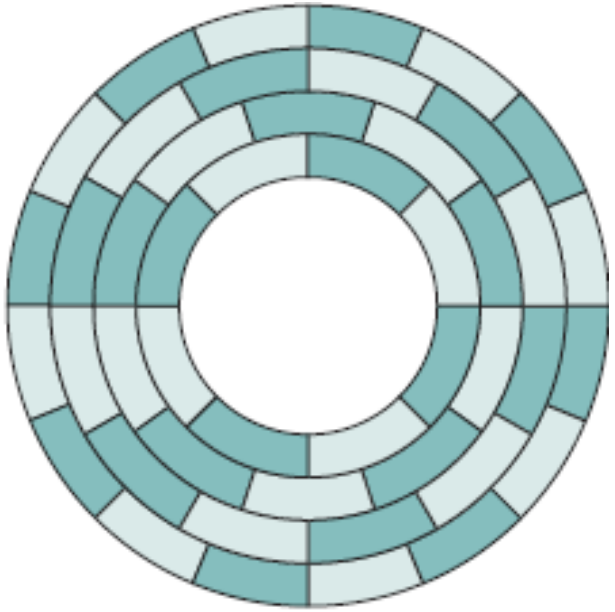
inner tracks – low data density



(a) Constant angular velocity

# Magnetic Disk – Data Organization

- To increase density, modern hard disk systems use a technique known as **multiple zone recording**
  - surface is divided into a number of concentric zones
  - within a zone, the number of bits per track is constant
  - Zones farther from the center contain more bits (more sectors)
- **Adv:** Greater overall storage capacity
- **Disadv:** Complex circuitry



(b) Multiple zoned recording

# Magnetic Disk - Performance

- **Seek time**
  - The time it takes to position the head at the desired track
  - Once on the track, the disk controller waits until the desired sector rotates to line up
- **Rotational Delay/Latency**
  - Time it takes for the beginning of the sector to reach the head
- **Access time**
  - Time it takes to get head into position for r/w operation
  - Seek time + Rotational delay
- **Transfer time**
  - Time required for the transfer (r/w) of a sector

# Magnetic Disk – Performance

- Transfer time, $$T = \frac{b}{rN}$$

  $T$ = transfer time

  $b$ = number of bytes to be transferred

  $N$ = number of bytes on a track

  $r$ = rotation speed, in revolutions per second

- Average access time, $$T_a = T_s + \frac{1}{2r} + \frac{b}{rN}$$

  $T_a$ = average access time

  $T_s$ = average seek time

- **Ex:** Consider a disk with average seek time of 4 ms, rotation speed of 15,000 rpm, and 512-byte sectors with 500 sectors per track. Estimate the total transfer time to read a file consisting of 2500 sectors for a total of 1.28 Mbytes.

# Exercises

- Consider a magnetic disk drive with 8 surfaces, 512 tracks per surface, and 64 sectors per track. Sector size is 1 kB. The average seek time is 8 ms, the track-to-track access time is 1.5 ms, and the drive rotates at 3600 rpm. Successive tracks in a cylinder can be read without head movement. Estimate the time required to transfer a 5-MB file for both the following types of file organization:
  - **Sequential file organization**: Assume this file is stored in successive sectors and tracks, starting at sector 0, and track 0.
  - **Random file organization**: Assume the file is stored in random sectors and tracks

# Optical memory

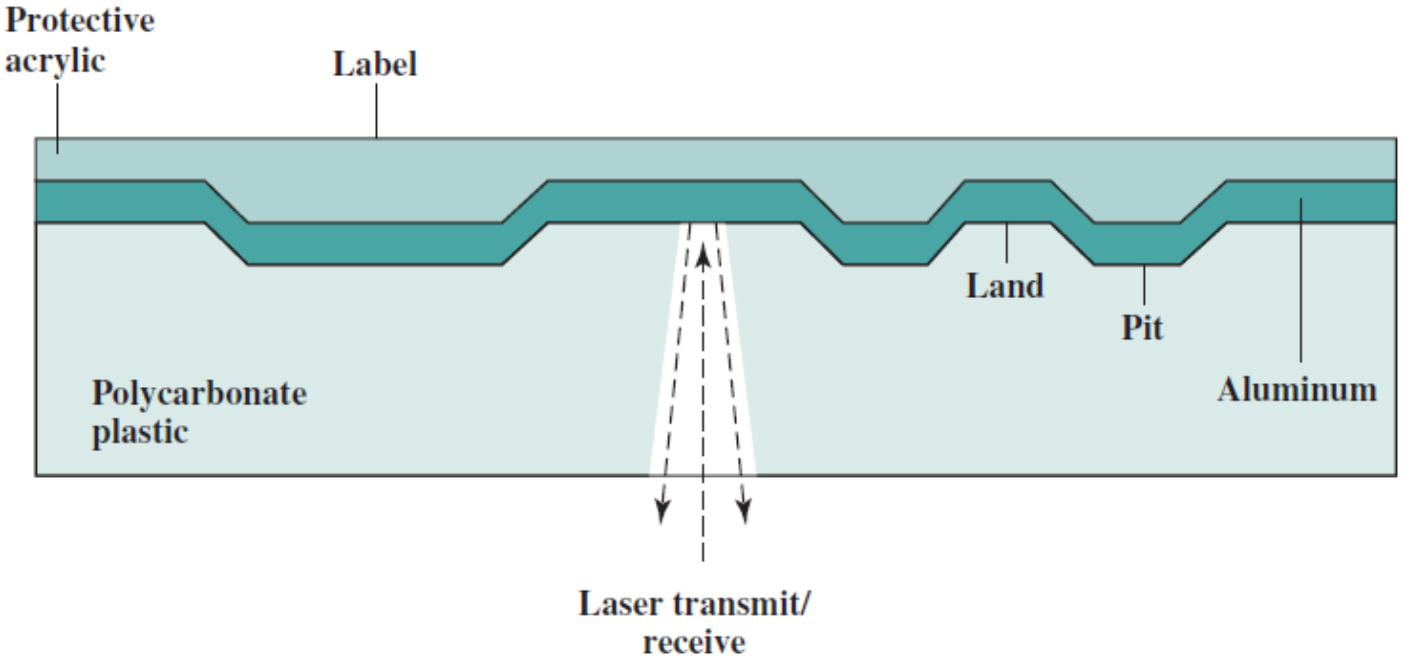
- Compact disk
  - Digital data is imprinted as a series of microscopic pits on the surface of the polycarbonate.
  - Done with a finely focused, high intensity laser
  - CD is read by a low-powered laser, while a motor spins the disk, intensity of the reflected light of the laser changes as it encounters a **pit.**

•The disk contains a single spiral track.

•Sectors near the outside of the disk are the same length as those near the inside. Information is packed evenly

•Scanned at the same rate by rotating the disk at a variable speed.

•The pits are then read by the laser at a **constant linear velocity (CLV)**.

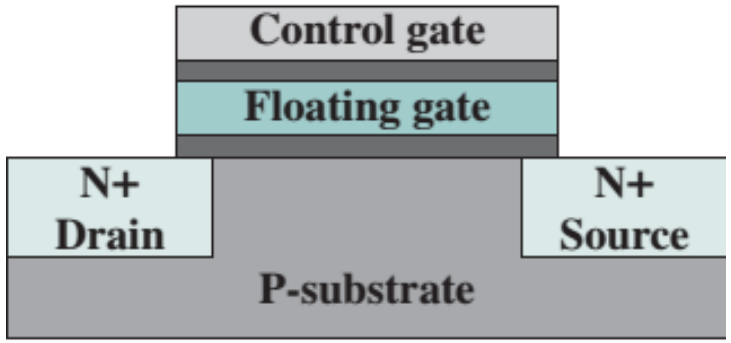


# Magnetic Tape

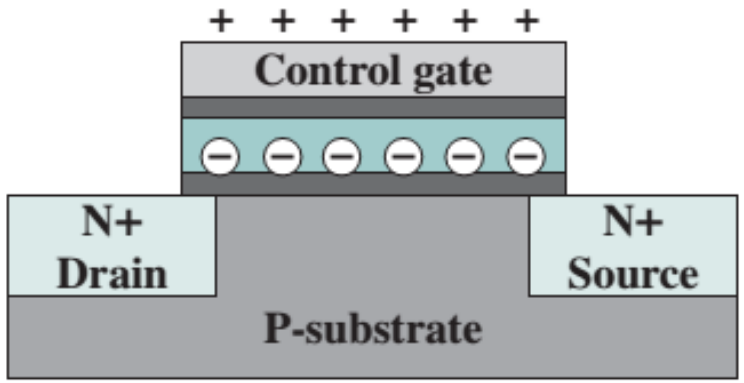
- Tape systems use the same reading and recording techniques as disk systems
- The medium is flexible polyester (similar to that used in some clothing) tape coated with magnetizable material
- Data on the tape are structured as a number of parallel tracks running lengthwise – parallel recording
- Data are laid out as a sequence of bits along each track, as is done with magnetic disks – serial recording

# Flash Memory

- A type of semiconductor memory
- Used in many consumer electronic products – smart phones, GPS devices, MP3 players, digital cameras, USB devices
- A second gate called a floating gate is added to the transistor.
  - Initially, the floating gate does not interfere with the operation of the transistor – represents binary 1
  - Applying a large voltage causes electrons to be trapped on the floating gate, where they remain even if the power is disconnected – represents binary 0




(b) Flash memory cell in one state




(c) Flash memory cell in zero state

# Flash Memory

- **NOR flash memory**
  - the basic unit of access is a bit
  - high-speed random access
  - read and write data to specific locations, and can reference and retrieve a single byte
  - Used for storing cell phone OS, and BIOS in Windows systems
- **NAND flash memory**,
  - the basic unit is 16 or 32 bits
  - higher bit density
  - does not provide a random-access external address bus so the data must be read on a blockwise basis
  - used in USB flash drives, memory cards, and in SSDs

# Solid State Drives

- **Solid state** refers to electronic circuitry built with semiconductors
- A **solid state drive (SSD)** is a memory device made with solid state components that can be used as a replacement to a hard disk drive – NAND flash memory used for SSD
- The cost of flash-based SSDs has dropped and the performance and bit density increased, SSDs have become increasingly competitive with HDDs.

# Solid State Drives

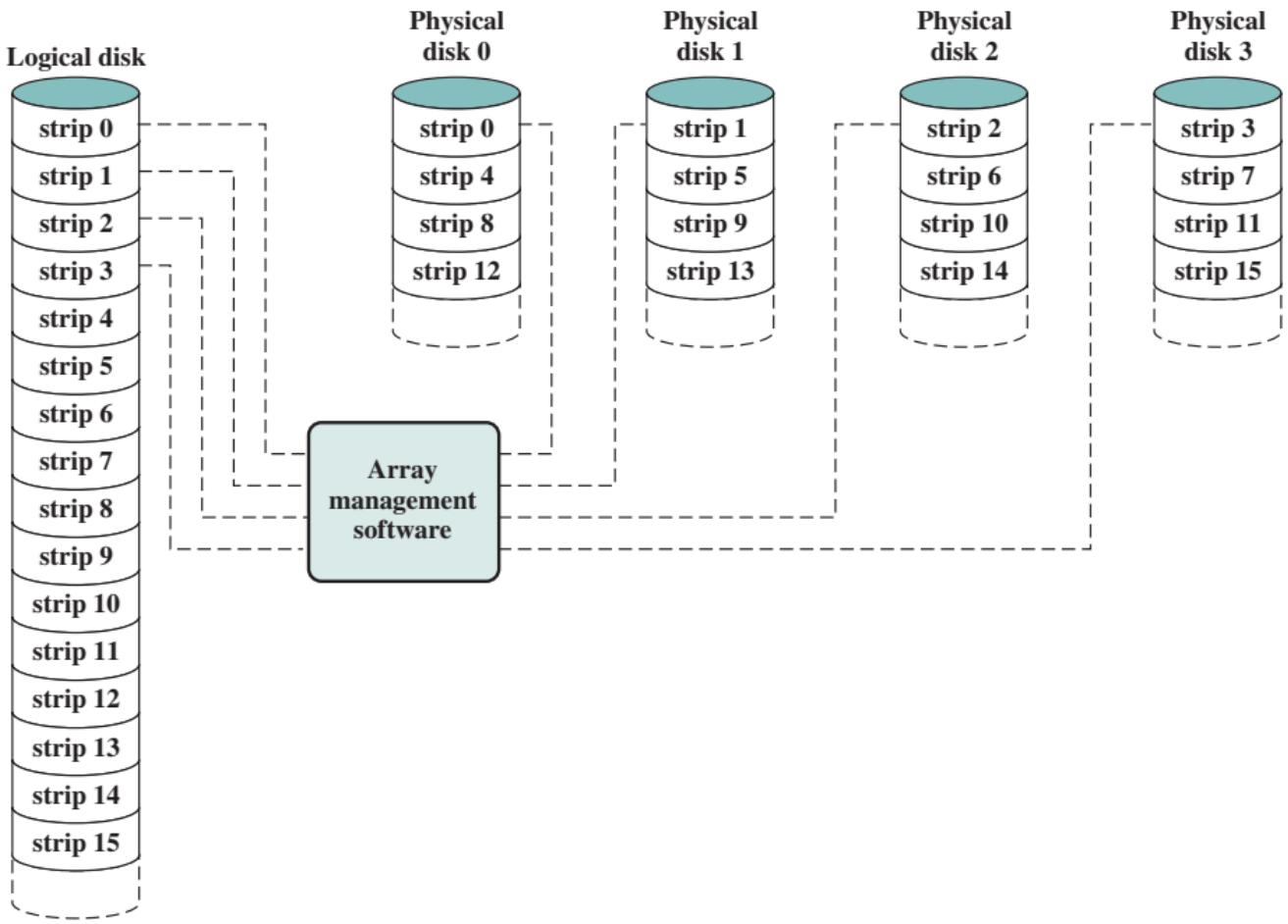
- Advantages of SSD over HDD
  - **High-performance input/output operations per second (IOPS):** Significantly increases performance I/O subsystems.
  - **Durability:** Less susceptible to physical shock and vibration.
  - **Longer lifespan:** SSDs are not susceptible to mechanical wear.
  - **Lower power consumption:** SSDs use as little as 2.1 watts of power per drive, considerably less than comparable-size HDDs.
  - **Quieter and cooler running capabilities:** Less floor space required, lower energy costs, and a greener enterprise.
  - **Lower access times and latency rates:** Over 10 times faster than the spinning disks in an HDD.
- Advantages of HDD over SSD
  - Low cost per bit and high capacity

# RAID

- RAID (Redundant Array of Independent Disks)
  - RAID is a set of physical disk drives viewed by the operating system as a single logical drive.
  - Data are distributed across the physical drives of an array in a scheme known as **striping**, described subsequently.
  - Redundant disk capacity is used to store parity information, which guarantees data recoverability in case of a disk failure.
- The RAID scheme consists of seven levels, zero through six.
- The details of the second and third characteristics differ for the different RAID levels. RAID 0 and RAID 1 do not support the third characteristic

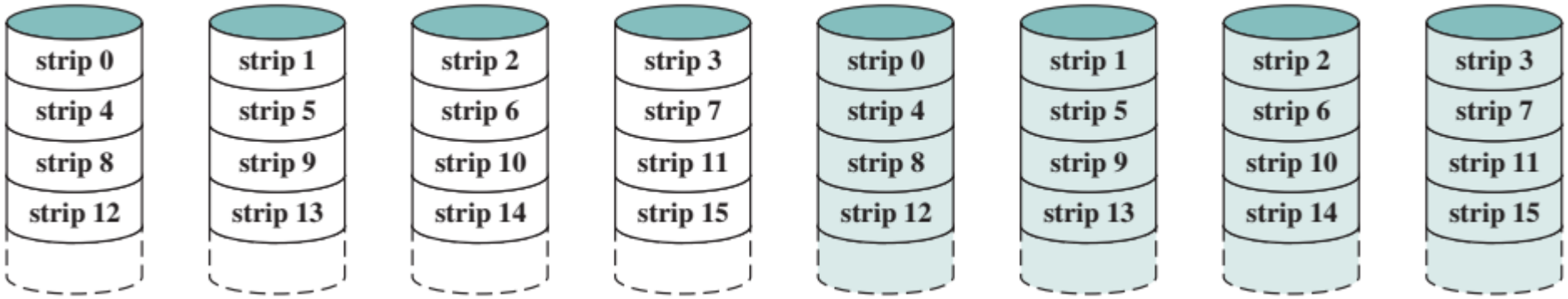
# RAID 0

- Data is striped across available drives
  - The logical disk is divided into strips; these **strips** may be physical blocks, sectors, or some other unit. The strips are mapped round robin to consecutive physical disks in the RAID array. A set of logically consecutive strips that maps exactly one strip to each array member is referred to as a **stripe**.
  - The advantage of this layout is that if a single I/O request consists of multiple logically contiguous strips, then up to n strips for that request can be handled in parallel, greatly reducing the I/O transfer time



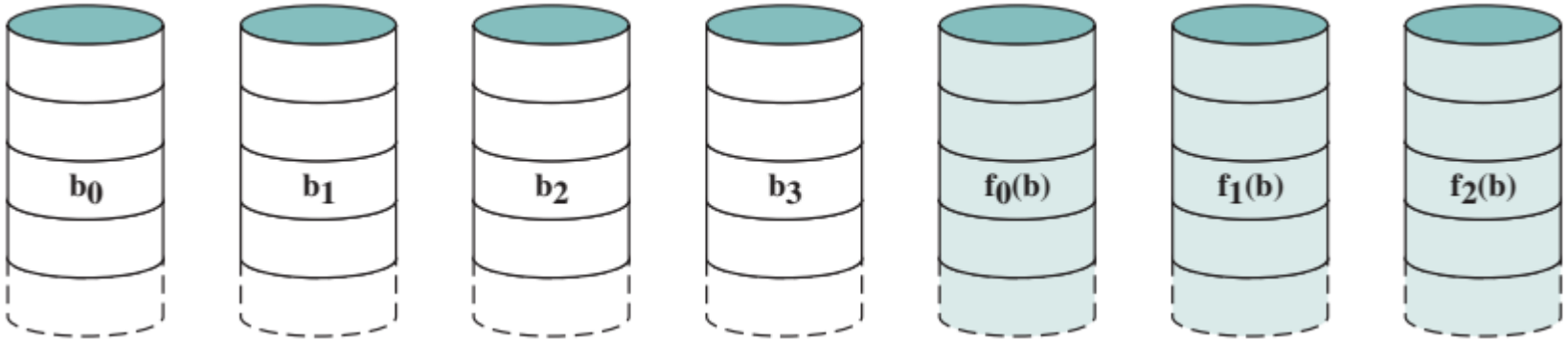
# RAID 1

- In RAID 1, redundancy is achieved by the simple expedient of duplicating all the data.
- Each logical strip is mapped to two separate physical disks so that every disk in the array has a mirror disk that contains the same data.
  - A read request can be serviced by either of the two disks that contains the requested data
  - A write request requires that both corresponding strips be updated, but this can be done in parallel
  - Recovery from a failure is simple. When a drive fails, the data may still be accessed from the second drive



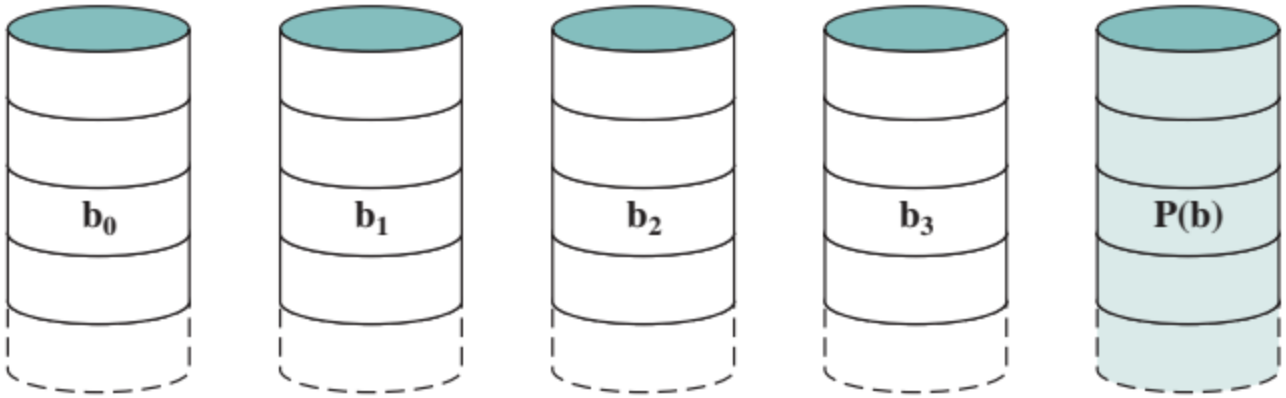
# RAID 2

- In RAID 2 the strips are very small, often as small as a single byte or word.
- An error-correcting code is calculated across corresponding bits on each data disk, and the bits of the code are stored in the corresponding bit positions on multiple parity disks.
  - In a read operation all disks are simultaneously accessed. The requested data and the associated error-correcting code are delivered to the array controller. If there is a single-bit error, the controller can recognize and correct the error instantly, so that the read access time is not slowed
  - In a write operation, all data disks and parity disks must be accessed for the write operation



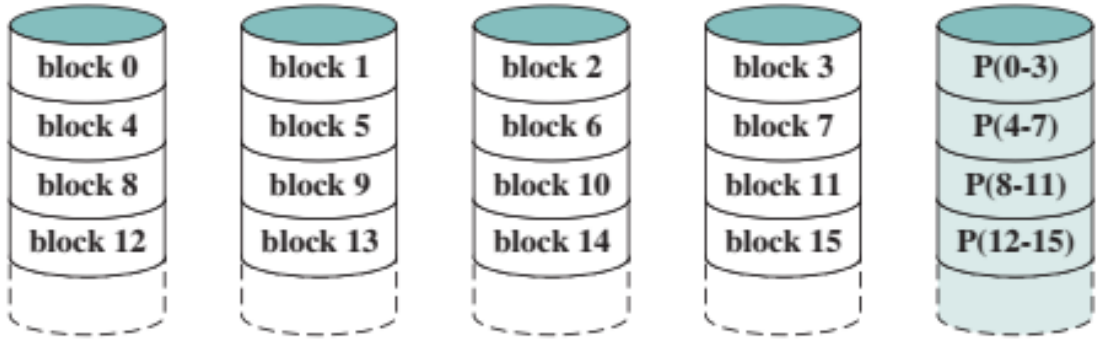
# RAID 3

- RAID 3 is organized similar to RAID 2. The difference is that RAID 3 requires only a single redundant disk, in which a simple parity bit is computed for the set of individual bits in the same position on all of the data disks.
- In the event of a drive failure, the parity drive is accessed and data is reconstructed from the remaining devices. Once the failed drive is replaced, the missing data can be restored on the new drive and operation resumed

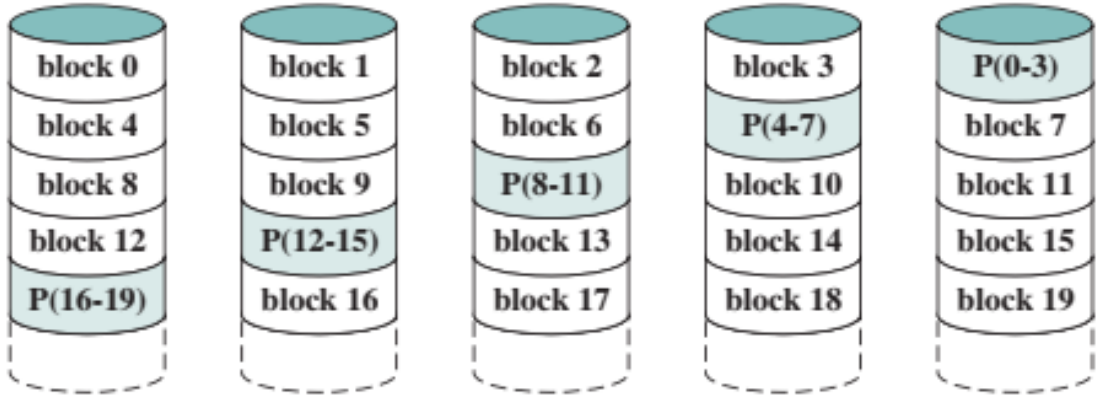


| Category | Level | Description | Disks Required | Data Availability |
|---|---|---|---|---|
| Striping | 0 | Nonredundant | $N$ | Lower than single disk |
| Mirroring | 1 | Mirrored | $2N$ | Higher than RAID 2, 3, 4, or 5; lower than RAID 6 |
| Parallel access | 2 | Redundant via Hamming code | $N + m$ | Much higher than single disk; comparable to RAID 3, 4, or 5 |
| | 3 | Bit-interleaved parity | $N + 1$ | Much higher than single disk; comparable to RAID 2, 4, or 5 |
| Independent access | 4 | Block-interleaved parity | $N + 1$ | Much higher than single disk; comparable to RAID 2, 3, or 5 |
| | 5 | Block-interleaved distributed parity | $N + 1$ | Much higher than single disk; comparable to RAID 2, 3, or 4 |
| | 6 | Block-interleaved dual distributed parity | $N + 2$ | Highest of all listed alternatives |

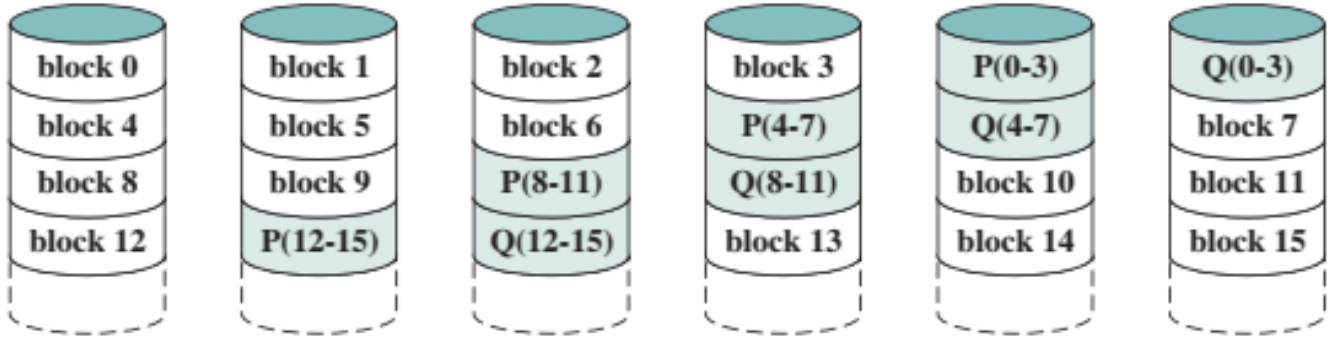
# RAID 4, 5, 6




(e) RAID 4 (Block-level parity)




(f) RAID 5 (Block-level distributed parity)




(g) RAID 6 (Dual redundancy)

*Students are encouraged to study the differences between all the RAID levels by themselves from the textbook*